ASSIGNMENT 02

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BCA (AI & DS) - C

1. What are epoch, batch size, and stratified?

An epoch is one complete pass of the full training dataset through the model.

Batch size is the number of samples processed before weights are updated. Small batches may generalize better, while large batches can be faster.

Stratified sampling keeps class proportions balanced in training and testing datasets.

These concepts are important for model training efficiency and evaluation.

2. Explain the different types of activation functions.

Sigmoid produces values between 0 and 1 and is useful in binary classification.

Tanh outputs between -1 and 1 and is zero-centered. ReLU outputs positive inputs directly and zero otherwise.

Leaky ReLU improves standard ReLU by allowing small gradients for negative values.

Softmax converts outputs into probability distributions for multi-class classification.

3. How we handle structured and unstructured data in ANNs and CNNs.

Structured data such as tabular datasets is commonly handled using ANNs after preprocessing and feature encoding.

Unstructured data such as images is effectively handled by CNNs using convolution layers for feature extraction.

CNNs can also process text and audio with suitable representations.

Choice of architecture depends on data type, dimensionality and problem requirements.

4. What is a layer, and define ANN and CNN layers?

A layer is a group of neurons that transforms inputs into outputs.

ANN includes input, hidden and output layers. Hidden layers perform computation and pattern learning.

CNN includes convolution layers, pooling layers and fully connected layers.

These layers together help extract features and perform classification.

5. What are the concepts used in ANN and CNN?

Important ANN concepts include neurons, weights, bias, activation functions, forward propagation and backpropagation.

CNN extends these with convolution, kernels, stride, padding, pooling and feature maps.

Both use loss functions, optimizers such as Adam or SGD, and regularization techniques like dropout.

Together these concepts allow networks to learn patterns and make accurate predictions.